Legged Robots Practical

26.09.2023

Practical Session Format

- First ~15 minutes will be a lecture on assignment
- Remaining time to work on assignment, ask questions, etc.
- Upload your code/reports on Moodle (1 per team)
- Make sure to check due dates!
- **All work must be your own**

Plan

Single Leg

- **W1:** Introduction
- **W2:** Derivation of double pendulum's kinematics and dynamics
- **W3:** Jacobian (Cartesian PD + Force Control)
- **W4:** Inverse Kinematics (compare with force control)
- **W5:** Single-leg hopping

Biped

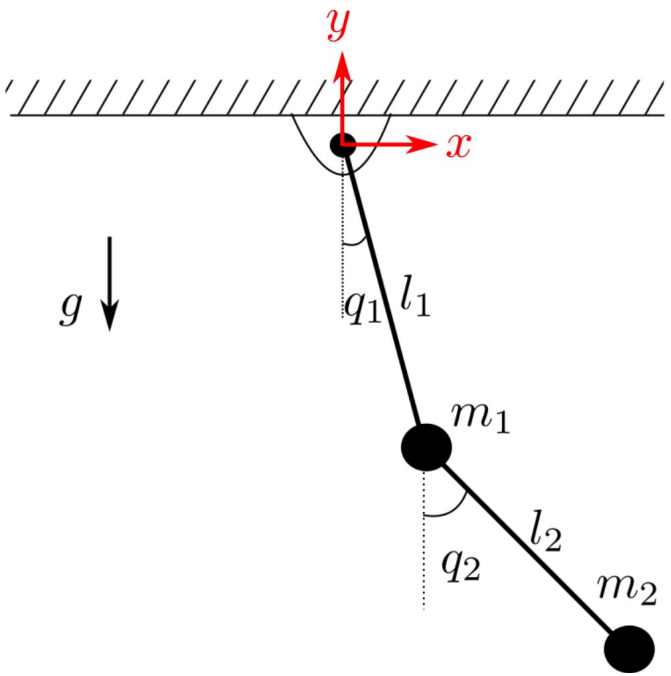
- **W6-8:** Biped Walking with Divergent Component of Motion (BD Atlas) **[Mini-Report]**

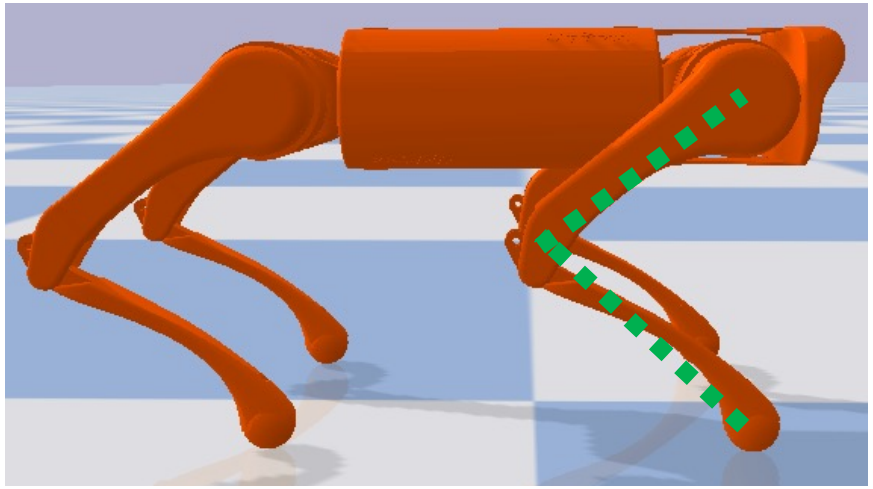
Quadruped

- **W9-10:** Quadruped CPG Trot
- **W11-14:** Quadruped Locomotion Project (CPGs, Deep RL) **[Report - Quadruped]**

Kinematics and Dynamics of the Double Pendulum

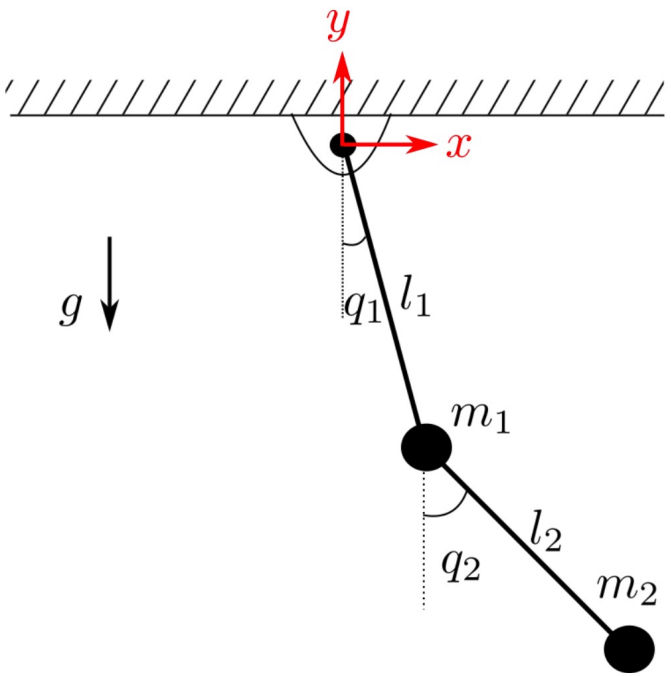
Why the double pendulum?

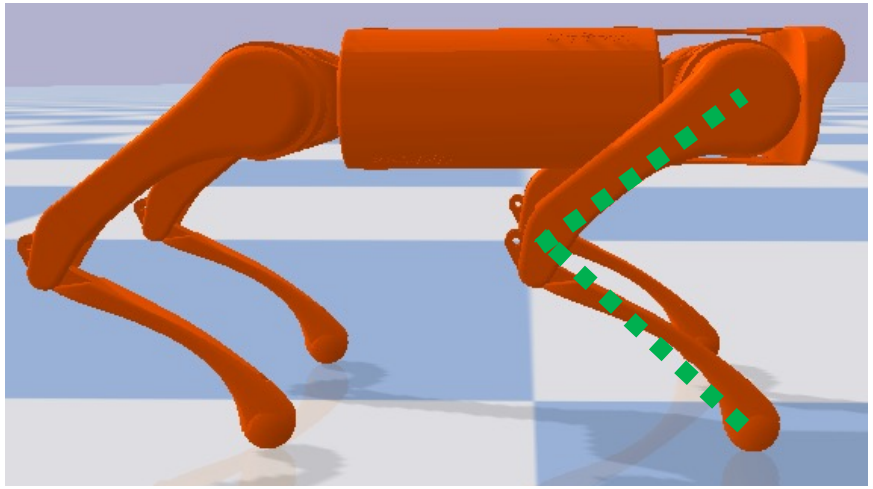




- Good approximation for a quadruped leg

Why derive the kinematics and dynamics?

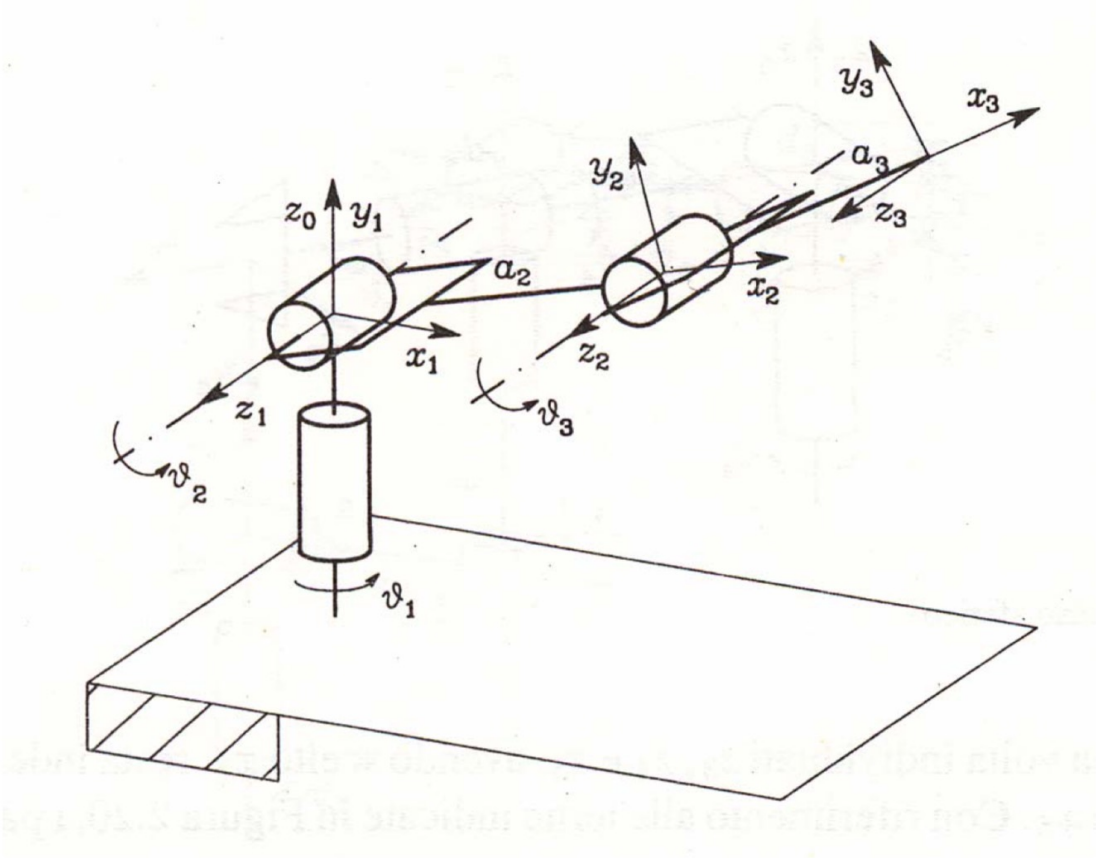




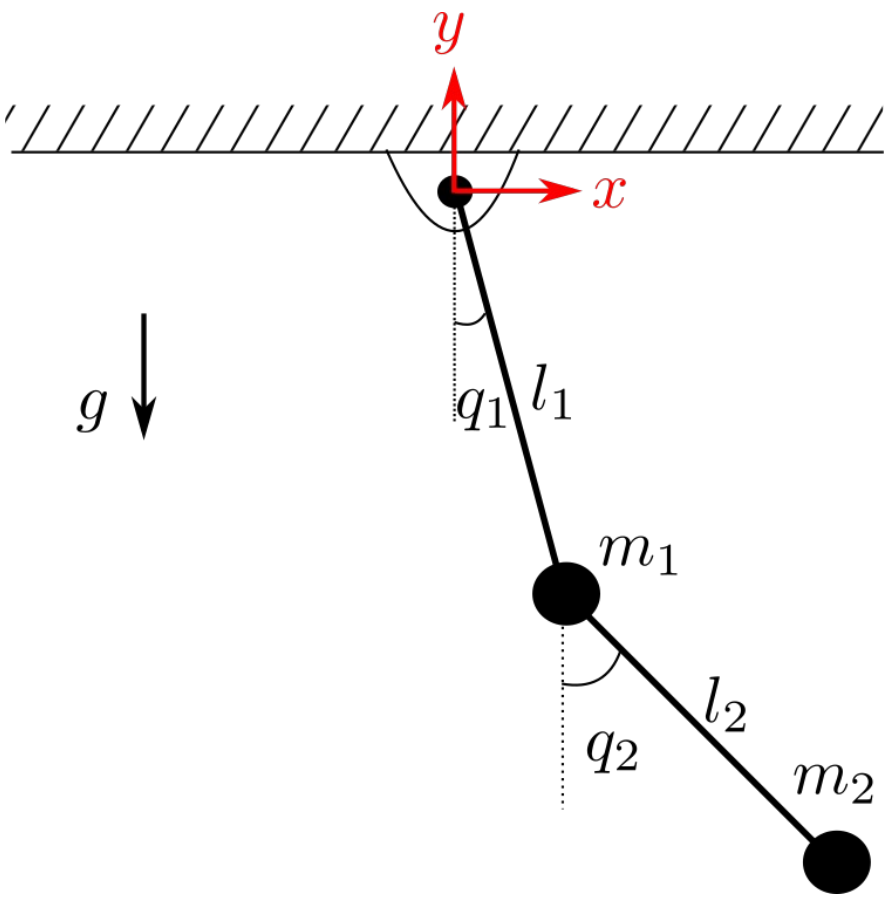
- To effectively control the system!
- The methodology will also be useful for other systems

Kinematics

- **Forward Kinematics**:
  - Input: joints' angles
  - Output: position and orientation of the end effector
- **Inverse Kinematics**:
  - Input: position and orientation of the end effector
  - Output: all the possible joints’ angles combinations



Forward Kinematics



Invert these relationships to obtain the inverse kinematics model (unique?)

$$x_1(q_1) = l_1 sin(q_1(t))$$

$$y_1(q_1) = -l_1 cos(q_1(t))$$

$$x_2(q_1, q_2) = l_1 sin(q_1(t)) + l_2 sin(q_2(t))$$

$$y_2(q_1, q_2) = -l_1 cos(q_1(t)) - l_2 cos(q_2(t))$$

$$\dot{x}_1(q_1, \dot{q}_1) = l_1 cos(q_1(t))\dot{q}_1(t) \qquad \dot{x} = \frac{dx}{dt}$$

$$\dot{y}_1(q_1, \dot{q}_1) = l_1 sin(q_1(t))\dot{q}_1(t)$$

$$\dot{x}_2(\dots) = l_1 cos(q_1(t))\dot{q}_1(t) + l_2 cos(q_2(t))\dot{q}_2(t)$$

$$\dot{y}_2(\dots) = l_1 sin(q_1(t))\dot{q}_1(t) + l_2 sin(q_2(t))\dot{q}_2(t)$$

Dynamics

- Dynamics studies the relation between the **joint actuator torques** and the **resulting motion**

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- **Inverse Dynamics (used for designing controllers):**

$$\boldsymbol{f} = \boldsymbol{m}\ddot{\boldsymbol{x}}$$

$$\boldsymbol{\tau} = \boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{G}(\boldsymbol{q}) + \boldsymbol{F}(\boldsymbol{q}, \dot{\boldsymbol{q}})$$

- **Direct (Forward) Dynamics (used for simulation)**

$$\ddot{\boldsymbol{x}} = \frac{\boldsymbol{f}}{\boldsymbol{m}}$$

$$\ddot{\boldsymbol{q}} = \boldsymbol{M}^{-1}(\boldsymbol{q})(\boldsymbol{\tau} - \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{G}(\boldsymbol{q}) + \boldsymbol{F}(\boldsymbol{q},\ \dot{\boldsymbol{q}}))$$

Equations of motion

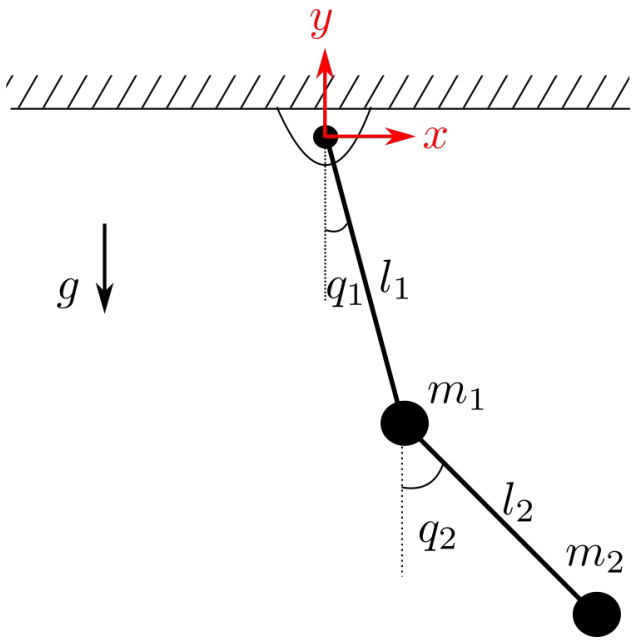
- **Lagrangian method:**
  - Variational method based on kinetics and potential energy

$$L(q, \dot{q}) = T(q, \dot{q}) - V(q)$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0$$

- **Newton-Euler recursive method**
  - Relies on $F = ma$ applied to each individual link of the robot

Dynamics



Potential Energy

$$\boldsymbol{q} = (\boldsymbol{q_1}, \boldsymbol{q_2})$$

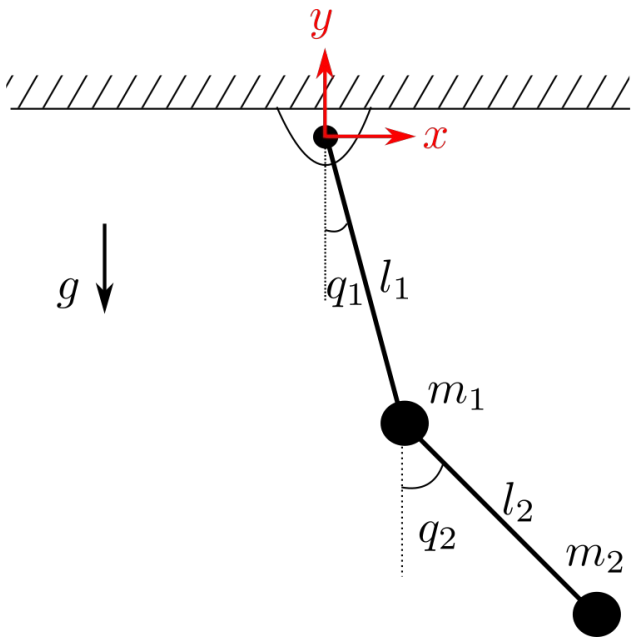
$$\boldsymbol{V}(\boldsymbol{q}) = \boldsymbol{V_1}(\boldsymbol{q}) + \boldsymbol{V_2}(\boldsymbol{q})$$

$$\boldsymbol{V_1}(q) = \boldsymbol{m_1 g y_1} = -\boldsymbol{m_1 g l_1 cos}(\boldsymbol{q_1})$$

$$\boldsymbol{V_2}(q) = \boldsymbol{m_2 g y_2} = \cdots$$

Potential energy is always a function of q and not $\dot{q}$!

Dynamics



Kinetic Energy

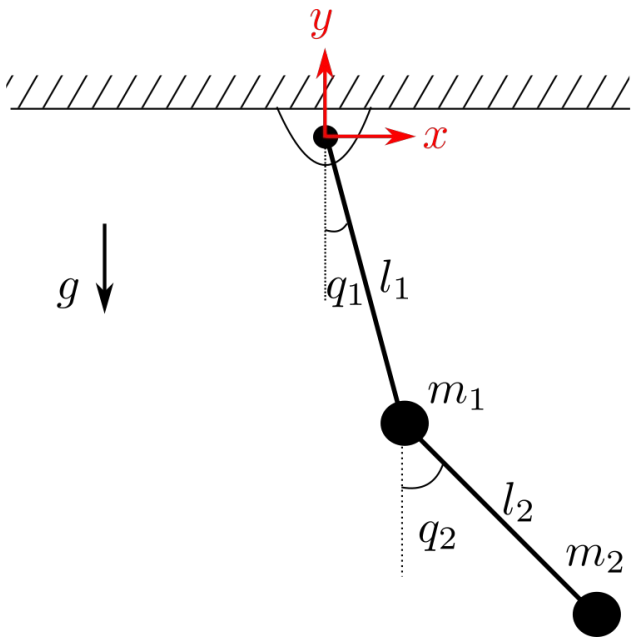
$$\boldsymbol{T(q,\dot{q}) = T_1(q,\dot{q}) + T_2(q,\dot{q})}$$

$$\boldsymbol{T_1(q,\dot{q}) = \frac{1}{2} m_1 \left(\dot{x}_1{}^2 + \dot{y}_1{}^2\right) = \frac{1}{2} m_1 l_1^2\, \dot{q}_1^2}$$

$$\boldsymbol{T_2(q,\dot{q}) = \frac{1}{2} m_2 \left(\dot{x}_2{}^2 + \dot{y}_2{}^2\right) = \cdots}$$

Kinetic energy is always a function of q and $\dot{q}$!

Dynamics



Equations of Motion

$$\frac{d}{dt}\left(\frac{\partial L(q,\dot{q})}{\partial \dot{q}_1}\right) - \frac{\partial L(q,\dot{q})}{\partial q_1} = 0$$

$$\frac{d}{dt}\left(\frac{\partial L(q,\dot{q})}{\partial \dot{q}_2}\right) - \frac{\partial L(q,\dot{q})}{\partial q_2} = 0$$

$$\Downarrow$$

$$\frac{d}{dt}\left(\frac{\partial T(q,\dot{q})}{\partial \dot{q}_1}\right) - \frac{\partial T(q,\dot{q})}{\partial q_1} + \frac{\partial V(q)}{\partial q_1} = 0$$

$$\frac{d}{dt}\left(\frac{\partial T(q,\dot{q})}{\partial \dot{q}_2}\right) - \frac{\partial T(q,\dot{q})}{\partial q_2} + \frac{\partial V(q)}{\partial q_2} = 0$$

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{\theta}_1}\right) - \frac{\partial T}{\partial \theta_1} + \frac{\partial V}{\partial \theta_1} = 0$$

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{\theta}_2}\right) - \frac{\partial T}{\partial \theta_2} + \frac{\partial V}{\partial \theta_2} = 0$$

Dynamics

$$\boldsymbol{q = (\theta_1, \theta_2)}$$

$$\frac{\partial T_1}{\partial \dot{\theta}_1} = m_1 l_1^2 \dot{\theta}_1$$

$$\frac{\partial T_2}{\partial \dot{\theta}_1} = m_2 l_1^2 \dot{\theta}_1 + m_2 l_1 l_2 \dot{\theta}_2 \cos(\theta_1 - \theta_2)$$

$$\frac{\partial T_1}{\partial \theta_1} = 0$$

$$\frac{\partial T_2}{\partial \theta_1} = -m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2)$$

$$\frac{\partial V_1}{\partial \theta_1} = m_1 g l_1 \sin(\theta_1)$$

$$\frac{\partial V_2}{\partial \theta_1} = m_2 g l_1 \sin(\theta_1)$$

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{\theta}_1}\right) - \frac{\partial T}{\partial \theta_1} + \frac{\partial V}{\partial \theta_1} = 0$$

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{\theta}_2}\right) - \frac{\partial T}{\partial \theta_2} + \frac{\partial V}{\partial \theta_2} = 0$$

Dynamics

$$\frac{\partial T_1}{\partial \dot{\theta}_2} = 0$$

$$\frac{\partial T_2}{\partial \dot{\theta}_2} = m_2 l_2^2 \dot{\theta}_2 + m_2 l_1 l_2 \dot{\theta}_1 \cos(\theta_1 - \theta_2)$$

$$\frac{\partial T_1}{\partial \theta_2} = 0$$

$$\frac{\partial T_2}{\partial \theta_2} = m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2)$$

$$\frac{\partial V_1}{\partial \theta_2} = 0$$

$$\frac{\partial V_2}{\partial \theta_2} = m_2 g l_2 \sin(\theta_1)$$

Dynamics

$$m_1 l_1^2 \ddot{\theta}_1 + m_2 l_1^2 \ddot{\theta}_1 + m_2 l_1 l_2 \cos(\theta_1 - \theta_2)\ddot{\theta}_2 - m_2 l_1 l_2 \dot{\theta}_2 (\dot{\theta}_1 - \dot{\theta}_2) \sin(\theta_1 - \theta_2)$$
$$+ m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) + m_1 g l_1 \sin(\theta_1) + m_2 g l_1 \sin(\theta_1) = 0$$

$$m_2 l_2^2 \ddot{\theta}_2 + m_2 l_1 l_2 \cos(\theta_1 - \theta_2)\ddot{\theta}_1 - m_2 l_1 l_2 \dot{\theta}_1 (\dot{\theta}_1 - \dot{\theta}_2) \sin(\theta_1 - \theta_2)$$
$$- m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) + m_2 g l_2 \sin(\theta_2) = 0$$

Dynamics

$$m_1 l_1^2 \ddot{\theta}_1 + m_2 l_1^2 \ddot{\theta}_1 + m_2 l_1 l_2 \cos(\theta_1 - \theta_2)\ddot{\theta}_2 - m_2 l_1 l_2 \dot{\theta}_2 (\dot{\theta}_1 - \dot{\theta}_2) \sin(\theta_1 - \theta_2)$$
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$$m_2 l_2^2 \ddot{\theta}_2 + m_2 l_1 l_2 \cos(\theta_1 - \theta_2)\ddot{\theta}_1 - m_2 l_1 l_2 \dot{\theta}_1 (\dot{\theta}_1 - \dot{\theta}_2) \sin(\theta_1 - \theta_2)$$
$$- m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) + m_2 g l_2 \sin(\theta_2) = 0$$

$$\tau = B(q)\ddot{q} + C(q, \dot{q}) + g(q)$$

Dynamics

$$m_1 l_1^2 \ddot{\theta}_1 + m_2 l_1^2 \ddot{\theta}_1 + m_2 l_1 l_2 \cos(\theta_1 - \theta_2)\ddot{\theta}_2 - m_2 l_1 l_2 \dot{\theta}_2 (\dot{\theta}_1 - \dot{\theta}_2) \sin(\theta_1 - \theta_2)$$
$$+ m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) + m_1 g l_1 \sin(\theta_1) + m_2 g l_1 \sin(\theta_1) = 0$$

$$m_2 l_2^2 \ddot{\theta}_2 + m_2 l_1 l_2 \cos(\theta_1 - \theta_2)\ddot{\theta}_1 - m_2 l_1 l_2 \dot{\theta}_1 (\dot{\theta}_1 - \dot{\theta}_2) \sin(\theta_1 - \theta_2)$$
$$- m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) + m_2 g l_2 \sin(\theta_2) = 0$$

$$\tau = B(q)\ddot{q} + C(q, \dot{q}) + g(q)$$

Dynamics

$$m_1 l_1^2 \ddot{\theta}_1 + m_2 l_1^2 \ddot{\theta}_1 + m_2 l_1 l_2 \cos(\theta_1 - \theta_2)\ddot{\theta}_2 - m_2 l_1 l_2 \dot{\theta}_2 (\dot{\theta}_1 - \dot{\theta}_2) \sin(\theta_1 - \theta_2) + m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) + m_1 g l_1 \sin(\theta_1) + m_2 g l_1 \sin(\theta_1) = 0$$

$$m_2 l_2^2 \ddot{\theta}_2 + m_2 l_1 l_2 \cos(\theta_1 - \theta_2)\ddot{\theta}_1 - m_2 l_1 l_2 \dot{\theta}_1 (\dot{\theta}_1 - \dot{\theta}_2) \sin(\theta_1 - \theta_2) - m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) + m_2 g l_2 \sin(\theta_2) = 0$$

$C(q, \dot{q})\, \dot{q}$

$$\tau = B(q)\ddot{q} + C(q, \dot{q}) + g(q)$$

Dynamics

$$m_1 l_1^2 \ddot{\theta}_1 + m_2 l_1^2 \ddot{\theta}_1 + m_2 l_1 l_2 \cos(\theta_1 - \theta_2)\ddot{\theta}_2 - m_2 l_1 l_2 \dot{\theta}_2 (\dot{\theta}_1 - \dot{\theta}_2) \sin(\theta_1 - \theta_2) + m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) + m_1 g l_1 \sin(\theta_1) + m_2 g l_1 \sin(\theta_1) = 0$$

$$m_2 l_2^2 \ddot{\theta}_2 + m_2 l_1 l_2 \cos(\theta_1 - \theta_2)\ddot{\theta}_1 - m_2 l_1 l_2 \dot{\theta}_1 (\dot{\theta}_1 - \dot{\theta}_2) \sin(\theta_1 - \theta_2) - m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) + m_2 g l_2 \sin(\theta_2) = 0$$

$C(q, \dot{q})\, \dot{q}$

$$\tau = B(q)\ddot{q} + C(q, \dot{q}) + g(q)$$

Derive the equations on your own

- Skeleton code on Moodle
- Upload your working code in Assignment 01 (1 per team)
- Today’s assignment is not graded